Deep Hedging under Black–Scholes and Heston Dynamics

Recover the known hedge first; then diagnose what changes under stochastic volatility

Micaela Marais · Glasson Osborne
Kaiden Wessels · Siphelele Nkosi

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University of Cape Town

Supervised by Prof. Ouwehand · MFE Research Project I

Can terminal-error training recover the known finite-grid hedge—and remain useful when volatility and the traded instrument set become more complex?

The hedge is learned through terminal portfolio error

$$\mathrm{HE} = \pi + \underbrace{\sum_{n=0}^{N-1} \delta_n \left(S_{t_{n+1}} - S_{t_n}\right)}_{\text{self-financing gains}} - (S_T - K)^+,$$

$$\min_{\pi,\theta} \ \mathbb{E}[\mathrm{HE}^2].$$

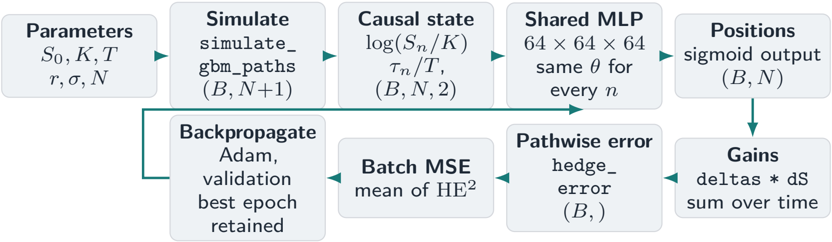
The analytic delta is never used as a training target.

- $N = 125$ discrete hedge dates in the baseline.
- $\delta_n = f_\theta(x_n)$ is adapted to information at t_n.
- Negative HE is a seller shortfall; reported seller loss is $L = -\mathrm{HE}$.
- With $r = 0$, the submitted experiments use undiscounted gains.

rebalance using current state

Source: Submitted report, Secs. 1.1, 2.2.2 and 3.1; Eqns. (1.1)–(1.2).

The implementation is differentiable end to end



Supervision is financial: payoff minus self-financing wealth.

Boundary: Function names and path construction are verified from report Appendix C; the separate executable notebook bundle is absent from this repository.

Source: Submitted report, Algorithm A.6.1 and Listings C.1–C.3. Shape contract follows the displayed array construction and shared network input.

Call economics guide architecture selection

$\log(S/K)$, τ/T → 64 tanh → 64 tanh → 64 tanh → 1 sigmoid

one shared map at all 125 dates

Candidate	Test RMSE	Time (s)
Shared norm. 64×3, tanh	**0.007891**	103.0
Shared norm. 64×2, ReLU	0.007912	93.2
Time-separated 16×1	0.008083	786.9

- **Log-moneyness** is scale-normalised and financially natural.
- **Weight sharing** exploits Markov and time-homogeneous structure.
- **Sigmoid** enforces the call-delta range $0 < \delta < 1$.
- A scalar premium is learned jointly with the hedge.

Selected on mean validation loss across three seeds; close alternatives remained.

Boundary: The model was selected, not uniquely dominant: the ReLU candidate won one seed.

Source: Submitted report, Secs. 3.1.2–3.1.3; Tables 3.1 and A.3–A.6.

The fair benchmark matches the grid and objective

No hedge
raw option risk

Black–Scholes delta
continuous-time rule sampled on the grid

Discrete-time MSE-optimal
conditional projection on the same grid

Degree-2 polynomial
deliberately low-capacity approximation

Neural hedge
terminal-MSE training without delta labels

same held-out paths

$$\delta_n^{\mathrm{DT}} = \frac{\mathrm{Cov}(g(S_T),\, \Delta S_n \mid S_{t_n})}{\mathrm{Var}(\Delta S_n \mid S_{t_n})}$$

The comparison asks whether the network recovers the finite-grid solution—not whether it beats correct Black–Scholes theory.

Source: Submitted report, Sec. 3.1.1 and Appendix A.4; Schweizer (1995, 2001).

The neural hedge recovers the finite-grid benchmark

Strategy	RMSE
No hedge	0.227 257
Black–Scholes delta	0.007 810
Discrete-time MSE-optimal	0.007 810
Degree-2 polynomial	0.022 183
Neural hedge	0.007 841

Boundary: These are final-benchmark values from an independent retraining, not architecture-search rows.

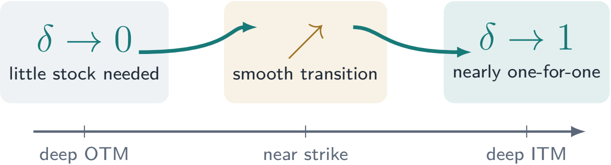
0.4%
neural RMSE above the discrete-time benchmark

0.163972
learned premium
vs. analytic 0.164111

Near equality is the validation result we wanted.

Source: Submitted report, Table 3.2 and Sec. 3.1.4.

The learned hedge has the right financial shape



- monotone in moneyness;
- close to both analytic curves;
- nearly indistinguishable on representative OTM, ATM and ITM paths.

The network learned the expected hedge function, not only a low aggregate loss.

Source: Economic reading of submitted report Figures 3.3 and 3.5; schematic above is not a data redraw.

Retraining preserves architecture quality

Retrained setting	NN RMSE	NN / DT
$K = 0.8$	0.006 656	1.082
$\sigma = 0.2$	0.003 680	1.082
$T = 1.0$	0.011 792	1.050
$N = 30$	0.015 595	1.009
$N = 250$	0.006 047	1.099

1.049

mean NN / discrete-time RMSE ratio across tested settings

Boundary: This is architecture reuse *with retraining*; it is not zero-shot generalisation.

Absolute error falls as the grid becomes finer, even when the relative gap rises because the benchmark improves faster.

Source: Submitted report, Table 4.1 and Sec. 4.1.1.

Missing information—not capacity—causes failure

K	σ	DT RMSE	NN RMSE
0.7	0.2	$0.000\,492$	$0.143\,314$
0.9	0.4	$0.007\,809$	$0.007\,841$
1.1	0.4	$0.009\,017$	$0.126\,961$
1.1	0.6	$0.013\,686$	$0.107\,178$

Model trained only at $K^\star = 0.9, \sigma^\star = 0.4$; evaluated without retraining.

Strike is hardcoded
Original moneyness uses $K^\star$.

Volatility is absent
The policy cannot condition on σ.

No parameter variation
The training distribution contains one scenario.

More paths from the same scenario cannot identify dependence on an unobserved, non-varying parameter.

Source: Submitted report, Table 4.2 and Sec. 4.1.2.

Conditioning restores cross-contract usefulness

$$\delta_n = f_\theta(\log(S_{t_n}/K),\ \tau_n/T,\ K,\ \sigma)$$

- Train once with $K \sim U(0.7, 1.1)$ and $\sigma \sim U(0.2, 0.6)$.
- Each path uses its analytic Black–Scholes premium; pricing is separated from hedging.
- Evaluation spans interpolation and moderate extrapolation without retraining.

Region	K	σ	DT	NN
Interp.	0.9	0.4	0.007 813	0.007 830
Interp.	1.1	0.6	0.013 715	0.013 783
Extrap.	0.6	0.4	0.002 220	0.002 241
Extrap.	0.9	0.1	0.001 019	0.003 404
Extrap.	1.3	0.8	0.019 187	0.020 058

Boundary: Interpolation is strong; volatility extrapolation is visibly harder.

Useful generalisation required both the right information set and a training distribution that varied those inputs.

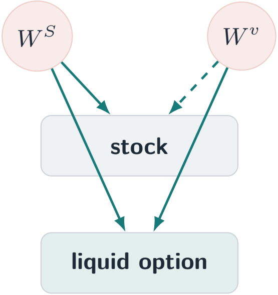
Source: Submitted report, Sec. 4.1.3, Table 4.3 and Algorithm A.6.3.

Heston introduces a second source of risk

$$
\begin{aligned}
dS_t &= rS_t\,dt + \sqrt{v_t}S_t\,dW_t^S,\\
dv_t &= \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v,\\
dW_t^S\,dW_t^v &= \rho\,dt.
\end{aligned}
$$

- v_t: stochastic instantaneous variance.
- κ, θ: mean reversion; ξ: volatility of volatility.
- $\rho = -0.70$: correlated spot–variance shocks in the submitted experiment.
- $2\kappa\theta = 0.64 \geq \xi^2 = 0.36$: Feller condition satisfied.

Source: Submitted report, Secs. 2.1.2 and 4.1.4; Heston (1993).



The hedge now chooses two traded positions

$$G_T = \sum_{n=0}^{N-1} \delta_n \, \Delta S_n + \sum_{n=0}^{N-1} \eta_n \, \Delta C_n^h$$

$$x_n = \left(\log \frac{S_n}{K}, \frac{T - t_n}{T}, \frac{v_n}{\theta}, \log \frac{S_n}{K_h}, \frac{T_h - t_n}{T_h}, \frac{C_n^h}{S_n} \right)$$

Target $K = 0.9, T = 0.5$; liquid option $K_h = 1, T_h = 1$, alive throughout.

Source: Submitted report, Sec. 4.1.4 and Algorithms A.3.3, A.6.10.

six causal state features

↓

shared fully connected network

↓

$$\delta_n = 5 \tanh z_{1,n}$$
$$\eta_n = 5 \tanh z_{2,n}$$

↓

two-instrument gains $\rightarrow$ terminal MSE

- Signed positions are now economically necessary.
- Full-truncation protects variance; all positions are causal.

The frozen-volatility proxy changed the market

$$C_t^h \approx C_{\mathrm{BS}}(S_t, K_h, T_h - t, \sqrt{v_t})$$

Insert the current Heston variance into a familiar Black–Scholes price.

Why it seemed reasonable

- cheap to evaluate at every path state;
- responsive to the current variance;
- produces a volatility-sensitive second instrument;
- preliminary neural RMSE 0.008198 was encouraging.

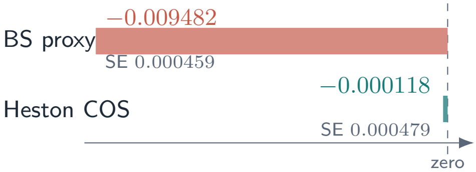
The hidden mismatch

- S_t, v_t follow Heston;
- C_t^h solves a frozen-volatility Black–Scholes equation.

Boundary: The proxy experiment motivates the refinement; it is not compared directly with the final COS experiment because the traded-option processes differ.

Source: Submitted report, Sec. 4.1.4 (pp. 24–25) and Algorithm A.6.5.

A martingale diagnostic exposes inconsistency



- Under $r = 0$, a consistently priced traded option should have approximately zero mean movement.
- The proxy displacement is large relative to its standard error.
- The COS residual is statistically negligible and consistent with discretisation.

The proxy is not merely imprecise in level; its path dynamics are inconsistent with the simulated Heston market.

Source: Submitted report, Table A.9 and Sec. 4.1.4. Bars reproduce reported values; zero is the risk-neutral $r = 0$ diagnostic reference.

Three omitted Heston terms explain the proxy drift

$$\mathrm{drift}(g) = \underbrace{\kappa(\theta - v)g_v}_{\text{mean reversion}\times\text{variance sensitivity}} + \underbrace{\tfrac{1}{2}\xi^2 v\, g_{vv}}_{\text{vol-of-vol convexity}} + \underbrace{\rho\xi v S\, g_{Sv}}_{\text{spot–variance interaction}} .$$

1

Apply Itô to $g(S, v, t) = C_{\mathrm{BS}}(S, K_h, T_h - t, \sqrt{v})$.

2

Cancel only the terms covered by the frozen-volatility BS PDE.

3

Three Heston terms remain, so drift is generally nonzero.

Boundary: The sign is not determined by ξ and ρ alone. Under the submitted parameters and simulated state distribution, the average residual was negative, matching the diagnostic.

Source: Submitted report, Sec. 4.1.4 and Appendix A.6.3.

COS pricing restores model consistency

Heston state (S_t, v_t, τ_h)

$\blacktriangledown$

Heston characteristic function

$\blacktriangledown$

256-term cosine expansion on $[-7, 7]$

$\blacktriangledown$

model-consistent liquid-option path C_t^h

5.1×10^{-8}

maximum COS–Carr–Madan price discrepancy

9.8×10^{-5}

finite-difference delta error

4.7×10^{-4}

finite-difference variance-sensitivity error

Source: Submitted report, Sec. 4.1.4 and Appendices A.6.1–A.6.2; Fang–Oosterlee (2008), Carr–Madan (1999). Validation is over the submitted traded range, not global.

The network beats the strongest tested heuristic

Representative strategy	RMSE	Loss CVaR95
No hedge	0.190 061	0.492 959
BS-proxy stock delta	0.019 997	0.046 360
Heston COS delta–vega	0.014 782	0.033 251
Stock-only NN (tanh)	0.018 166	0.041 379
Strongest tested heuristic	0.009 874	0.023 130
Stock + option NN	0.007 426	0.016 624

Seed	NN RMSE	Improvement %
111	0.007 806	20.6
222	0.007 715	21.8
333	0.007 346	25.5

Boundary: All Heston headline metrics use strategy-by-strategy fair-premium centering on the same test paths.

22.7%
mean RMSE improvement

27.1%
mean Loss CVaR95 improvement

The network won all three submitted pairwise seed comparisons against the strongest analytic heuristic tested.

Source: Submitted report, Tables 4.5–4.6 and A.11. Representative values and multi-seed means are deliberately separated.

The evidence is strong within explicit boundaries

What the evidence establishes

1. **Black–Scholes:** recovered the known finite-grid hedge.
2. **Conditioning:** repaired the identified information failure.
3. **Heston:** reduced terminal risk against the strongest tested heuristic once the liquid option was priced consistently.

Learned option positions broadly follow variance-hedging direction, but soften at high variance under terminal MSE.

Where the conclusion stops

- three seeds and evaluation-path premium centering;
- comparators are local heuristics, not a derived finite-grid multi-instrument optimum;
- frictionless simulated Heston market;
- full-information headline; observable-market-information robustness is not stock-return-only filtering.

A constrained, rigorously benchmarked neural policy can recover a known solution and adapt when the state and traded instrument set become more complex.

Source: Submitted report, Secs. 4.1.4 and 5; Figure 4.3 interpretation and stated limitations.

Backup: method-to-code map

Mathematical concept	Displayed implementation	Report evidence	Primary expert
Exact BS path simulation	`simulate_gbm_paths`	Listing C.1; Alg. A.3.1	Micaela / Glasson
Self-financing terminal error	`hedge_error`	Listing C.2; Alg. A.6.1	Micaela / Glasson
Shared Markov hedge	`SharedMarkovHedge`	Listing C.3; Sec. 2.2.2	Kaiden / Micaela
Heston full-truncation simulation	PyTorch loop in Listing C.4	Listing C.4; Alg. A.3.3	Glasson / Siphelele
COS liquid-option pricing	Algorithms A.6.6–A.6.8	App. A.6.1–A.6.2	Glasson
Two-instrument neural gain	Algorithm A.6.10	Sec. 4.1.4	Glasson

Boundary: The submitted report and its displayed code are present; notebook filenames, result CSV paths and `requirements.txt` are absent from this repository. See `OPEN_QUESTIONS.md`.

Backup: tensor shapes

BS paths $(B, N+1)$ → state stack $(B, N, 2)$ → shared calls $(B \cdot N, 2)$ → deltas (B, N) → terminal error $(B,)$

Heston S, v, C^h each $(B, N+1)$ → state stack $(B, N, 6)$ → shared calls $(B \cdot N, 6)$ → (δ, η) $(B, N, 2)$ → gain / error $(B,)$

BS path allocation and two-feature input are explicit in Listings C.1 and C.3. Heston array indexing and the six-feature/two-output contract are explicit in Listing C.4 and Algorithm A.6.10; flattening is the shared batched evaluation contract described in the report.

Backup: the discrete-time conditional projection

$$\min_{\pi,\delta} \quad \mathbb{E}\left[\left(\pi + \sum_{j=0}^{N-1} \delta_j \Delta S_j - H\right)^2\right],$$

$$0 = \mathbb{E}\left[\left(H - \pi - \sum_{j=0}^{N-1} \delta_j \Delta S_j\right) \Delta S_n \mid \mathcal{F}_{t_n}\right].$$

Under the submitted $r = 0$ risk-neutral simulation, S is a martingale, so

$$\delta_n^{\mathrm{DT}} = \frac{\mathbb{E}[H \Delta S_n \mid \mathcal{F}_{t_n}]}{\mathbb{E}[(\Delta S_n)^2 \mid \mathcal{F}_{t_n}]} = \frac{\mathrm{Cov}(H, \Delta S_n \mid S_{t_n})}{\mathrm{Var}(\Delta S_n \mid S_{t_n})}.$$

Boundary: This simplification is tied to the submitted $r = 0$ martingale setting; local and global quadratic hedges need not coincide more generally.

Source: Submitted report, Appendix A.4; Schweizer (1995, 2001).

Backup: full frozen-volatility drift derivation

For $g(S,v,t) = C_{\mathrm{BS}}(S, K_h, T_h - t, \sqrt{v})$, Itô's lemma under Heston gives

$$dg = \Big[g_t + rSg_S + \kappa(\theta - v)g_v + \tfrac{1}{2}vS^2 g_{SS} + \tfrac{1}{2}\xi^2 v g_{vv} + \rho\xi v S g_{Sv}\Big]dt \\ + g_S\sqrt{v}S\,dW^S + g_v\xi\sqrt{v}\,dW^v.$$

The frozen-v Black–Scholes PDE cancels only

$$g_t + rSg_S + \tfrac{1}{2}vS^2 g_{SS} - rg = 0.$$

Hence the discounted residual is

$$\kappa(\theta - v)g_v + \tfrac{1}{2}\xi^2 v g_{vv} + \rho\xi v S g_{Sv},$$

and with submitted $r = 0$ this is also the undiscounted drift residual.

Source: Submitted report, Appendix A.6.3.

Backup: COS pricing in one equation

$$C_{\text{COS}} \approx e^{-r\tau} \sum_{k=0}^{N_{\text{COS}}-1}{}' A_k V_k,$$

$$A_k = \frac{2}{b-a} \Re\left[\varphi(u_k) e^{-iu_k a}\right], \qquad u_k = \frac{k\pi}{b-a}.$$

A_k approximates the density coefficient; V_k is the analytic call-payoff coefficient.

Source: Submitted report, Algorithms A.6.6–A.6.7 and Appendix A.6.2.

- Heston characteristic function φ.
- $N_{\text{COS}} = 256$, $[a, b] = [-7, 7]$.
- Traded validation range: $\tau_h \in [0.5, 1]$.
- Carr–Madan check: $\alpha = 1.5$, $u \in [0, 200]$, 20,000 points.

Boundary: Accuracy is validated on the traded grid, not claimed globally.

Backup: Heston simulation code

```
for n in range(N_STEPS):
    z1 = torch.randn(n_paths, device=device)
    z2 = torch.randn(n_paths, device=device)
    z_s = z1
    z_v = RHO*z1 + math.sqrt(max(1.0-RHO**2, 0.0))*z2
    v_pos = torch.clamp(v[:, n], min=0.0)
    S[:, n+1] = S[:, n] * torch.exp(
        (R - 0.5*v_pos)*dt + torch.sqrt(v_pos)*sqrt_dt*z_s
    )
    v_next = v[:, n] + KAPPA*(THETA-v_pos)*dt \
            + XI*torch.sqrt(v_pos)*sqrt_dt*z_v
    v[:, n+1] = torch.clamp(v_next, min=0.0)
```

Source: Adapted only for line length from submitted report Listing C.4. Purpose: correlated Heston shocks, log-Euler stock update and full-truncation variance update. Primary speaker: Glasson / Siphelele.

Backup: two-instrument gain construction

```
# state at t_n only
x_n = [log(S_n/K), (T-t_n)/T, v_n/THETA,
      log(S_n/K_H), (T_H-t_n)/T_H, C_h_n/S_n]
z_stock, z_option = shared_mlp(x_n)
delta_n = 5.0 * tanh(z_stock)
eta_n = 5.0 * tanh(z_option)

gain = sum(delta_n * (S_next-S_n)
          + eta_n * (C_h_next-C_h_n), over="time")
hedge_error = premium + gain - maximum(S_T-K, 0.0)
loss = mean(hedge_error**2)
```

Boundary: This is a faithful pseudocode rendering of Algorithm A.6.10, not a verbatim excerpt from the absent executable notebook.

Source: Submitted report, Sec. 4.1.4 and Algorithm A.6.10. Primary speaker: Glasson.

Backup: data separation and centering

Training
fit weights and learned premiu

Validation
select architecture / best epoch

Test
one final risk evaluation

independent paths

independent paths

independent paths

Boundary: Heston headline fair premiums are estimated on the same test paths as the risk metrics. Interpret them as test-sample-centred hedge-quality comparisons, not fully out-of-sample pricing results.

Source: Submitted report, Tables A.1–A.2 and Algorithms A.6.1–A.6.2.

Backup: three-seed Heston comparison

Seed	NN RMSE	Heuristic RMSE	RMSE imp. (%)	CVaR95 imp. (%)
111	0.007 806	0.009 838	20.6	26.3
222	0.007 715	0.009 871	21.8	27.0
333	0.007 346	0.009 864	25.5	28.0
		Mean pairwise improvement	22.7	27.1

Boundary: Three seeds demonstrate consistency across the submitted runs; they do not support high-powered confidence intervals.

Source: Submitted report, Table A.11.

Backup: position and information diagnostics

Position-bound check

- Network output range: $[-5, 5]$.
- Observed max $|\eta^{NN}| \approx 1.33$.
- COS delta–vega RMSE:
  - bound $[-5, 5]$: 0.014782;
  - bound $[-20, 20]$: 0.014667.

Observable-information check

- EWMA decay $\lambda = 0.94$.
- $\mathrm{corr}(\hat{v}_t, v_t) = 0.777613$.
- Relative variance-proxy RMSE $= 0.369375$.
- Neural mean RMSE 0.007877 vs heuristic 0.010734.

Boundary: C_t^h / S_t remains observable and variance-informative; this is not stock-return-only filtering.

Source: Submitted report, Sec. 4.1.4 and Tables A.10, A.12, A.15.

Backup: transaction costs change the learned policy

$$V_T^{\lambda} = \pi + \sum_{n=0}^{N-1} \delta_n \Delta S_n - \lambda \sum_{n=0}^{N-1} S_{t_n} |\delta_n - \delta_{n-1}|, \qquad \delta_{-1} = 0.$$

λ	Strategy	RMSE	Loss CVaR95	Turnover
0.0025	BS cost-adjusted	0.008 492	0.021 464	4.497 364
0.0025	Cost-aware NN	0.008 077	0.018 694	4.386 880
0.0050	BS cost-adjusted	0.010 013	0.025 565	4.497 364
0.0050	Cost-aware NN	0.008 331	0.019 269	4.284 602

Boundary: Comparator is a cost-adjusted frictionless delta, not a full optimal-cost benchmark such as Leland or Whalley–Wilmott.

Source: Submitted report, Appendix A.7 and Table A.16.

Backup: data-generation comparison

Paths	Method	RMSE	Loss CVaR95
10,000	Crude MC	0.009 430	0.021 266
10,000	LHS	0.009 201	0.020 845
10,000	Sobol + Brownian bridge	0.009 174	0.020 865
100,000	Crude MC	0.007 865	0.018 218
100,000	LHS	0.007 846	0.018 331
100,000	Sobol + Brownian bridge	0.007 840	0.018 125

At 10,000 paths, structured methods reduce RMSE by roughly 2.4–2.7%; by 100,000 paths the difference is negligible.

Source: Submitted report, Sec. 3.1.5 and Tables A.7–A.8.

Backup: experimental protocol

Experiment	Train	Val	Test	Batch	Training / seeds
Architecture selection	100k	20k	50k	256	60 epochs; 2026–2028
Final BS benchmark	100k	30k	100k	4096	250 epochs; base 123
Parameter robustness	50k	15k	50k	4096	150 epochs
Conditioned hedger	200k	30k	50k / case	4096	250 epochs; 777
Heston stock + option	100k	25k	100k	4096	150 epochs; base 123

- Adam learning rate 10^{-3} unless otherwise stated.
- Baseline: $S_0 = 1, K = 0.9, T = 0.5, r = 0, \sigma = 0.4, N = 125.$
- Architecture search uses more updates per epoch via batch size 256; final experiments use 4096.

Source: Submitted report, Appendix A.1, Table A.1.

Backup: AI assistance and verification

Assistance area disclosed	Human verification reported
Code drafting and debugging	Executed notebooks, code inspection and exported results
Mathematical explanations and LaTeX	Checked against theory and cited sources
Report organisation and consistency	Tables checked against CSV/TeX outputs and figures
COS implementation support	Independent Carr–Madan Fourier price check
Neural experiment support	BS analytic and finite-grid benchmarks; archived outputs
Documentation and reproducibility	Checklists, packaging review and author decisions

Boundary: The report names ChatGPT and Claude. It states that AI suggestions were not authoritative and that all final methodological and interpretive decisions remained with the authors.

Source: Submitted report, Appendix B.

Backup: architecture selection by seed

Seed	Validation winner	Val. loss	Test RMSE
2026	Shared norm. 64×3, tanh	6.025×10^{-5}	$0.007\,883$
2027	Shared norm. 64×3, tanh	6.164×10^{-5}	$0.007\,916$
2028	Shared norm. 64×2, ReLU	6.014×10^{-5}	$0.007\,855$

The 64×3 tanh model won two seeds and had the best mean validation loss and test RMSE; the ReLU alternative was close.

Source: Submitted report, Tables A.3–A.6.

Backup: references I

Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.

Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. Deep hedging. *Quantitative Finance*, 19(8):1271–1291, 2019.

Peter Carr and Dilip B. Madan. Option valuation using the fast fourier transform. *Journal of Computational Finance*, 2(4):61–73, 1999.

Fang Fang and Cornelis W. Oosterlee. A novel pricing method for european options based on fourier-cosine series expansions. *SIAM Journal on Scientific Computing*, 31(2):826–848, 2008.

Hans Föllmer and Martin Schweizer. Hedging of contingent claims under incomplete information. In M. H. A. Davis and R. J. Elliott, editors, *Applied Stochastic Analysis*, pages 389–414. Gordon and Breach, 1991.

Jim Gatheral. *The Volatility Surface: A Practitioner's Guide*. Wiley, 2006.

Paul Glasserman. *Monte Carlo Methods in Financial Engineering*. Springer, 2004.

Steven L. Heston. A closed-form solution for options with stochastic volatility. *The Review of Financial Studies*, 6(2):327–343, 1993.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *International Conference on Learning Representations*, 2015.

Hayne E. Leland. Option pricing and replication with transactions costs. *Journal of Finance*, 40(5):1283–1301, 1985.

Roger Lord, Remmert Koekkoek, and Dick van Dijk. A comparison of biased simulation schemes for stochastic volatility models. *Quantitative Finance*, 10(2):177–194, 2010.

Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

Johannes Ruf and Weiguan Wang. Neural networks for option pricing and hedging: A literature review. *Journal of Computational Finance*, 24(1):1–46, 2020.

Martin Schweizer. Variance-optimal hedging in discrete time. *Mathematics of Operations Research*, 20(1):1–32, 1995.

Martin Schweizer. A guided tour through quadratic hedging approaches. In Elyes Jouini, Jakša Cvitanić, and Marek Musiela, editors, *Option Pricing, Interest Rates and Risk Management*, pages 538–574. Cambridge University Press, 2001.

A. E. Whalley and P. Wilmott. An asymptotic analysis of an optimal hedging model for option pricing with transaction costs. *Mathematical Finance*, 7(3):307–324, 1997.